

Hongchen (Steven) Yang

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Education

Brandeis University, Waltham, MA

Expected Dec 2026

B.S. in Computer Science GPA: Major 4.0 / 4.0; Overall 3.765 / 4.0

Dean's List (All Semesters); Provost's Summer Undergraduate Research Fellowship (Summer 2025)

Relevant Coursework: Computer Systems Design; Database Management; Operating Systems; Machine Learning; Theory of Computation

Research Experience

Smart & Scalable Data Systems (SSD) Lab, Researcher — Brandeis University

May 2025 – Present

- Survey recent LSM-tree and NoSQL systems literature to study how shifting workloads, skew, and burstiness affect compaction behavior and performance.
- Categorize RocksDB tunable knobs by objectives such as throughput, latency, CPU utilization, and amplification to identify the most impactful parameters for optimization.
- Reduce configuration search cost by grouping dependent parameters and pruning knobs that can be derived from others, making large search spaces more tractable.
- Investigate workload-aware knob selection using Bayesian optimization, tree-based surrogate models, and lightweight online learning to adapt parameters under changing workloads.
- Explore RocksDB source code and build prototype benchmarking workflows to evaluate tuning policies, informing the design of *FluidLSM*, a self-adaptive LSM-based key-value store.

Data Science Intern (Independent Study) — Brandeis University

Jan 2026 – Present

- Design a benchmark for fragmented higher-education administrative data to measure how text-to-SQL systems degrade under schema heterogeneity, missingness, and record fragmentation.
- Redesign synthetic schemas, data generators, and controllable fragmentation knobs so baseline and fragmented datasets can be produced from a shared source for controlled experiments.
- Define fragmentation metrics based on query execution quality, join efficiency, and text-to-SQL task success to quantify how structural changes affect downstream performance.
- Develop Python ETL and evaluation pipelines for provenance tracking, fragmentation analysis, and baseline vs. fragmented text-to-SQL comparisons.

Teaching Experience

Teaching Assistant, Introduction to Database — Brandeis University

Jan 2026 – Present

- Lead office hours and review sessions for about 50 students covering SQL, ER modeling, normalization, indexing, and query optimization.
- Grade assignments and exams and provide feedback on query design, query optimization, relational modeling, and database system fundamentals.

Internship Experience

Shanghai Development Center of Computer Software Technology — Shanghai, China

Software Engineering Intern

May 2024 – Aug 2024

- Traced Supersonic AI's text-to-SQL workflow from user prompt to generated query to understand translation, parsing, planning, and execution stages.
- Studied the backend architecture of the RuoYi framework and its Maven-based integration with Spring Boot, MyBatis, and Redis.
- Used Linux based development workflows to run services, inspect logs, and support debugging during backend development tasks.
- Applied Java streams and lambda expressions to refactor service logic and improve code readability and maintainability.
- Investigated how data structures and algorithmic choices influence backend performance, strengthening practical understanding of system optimization.

Projects

Blacklight Privacy Detection Tool — The Markup & Brandeis University

Aug 2025 – Dec 2025

[Blacklight Update Blog](#)

- Contributed to the research and development of Blacklight updates that detect TikTok and X tracking pixels on websites.
- Worked with a Brandeis capstone team and The Markup product team on a public facing privacy inspection tool used for more than 18 million scans.
- Supported feature development and validation for additional third-party pixel detection to improve tracker visibility for end users.
- Collaborated across research and product contexts to translate privacy measurement work into a deployed public interest auditing tool.

Technical Skills

Languages: Java, C++, Python, Go, Rust, JavaScript, TypeScript, SQL

Frameworks & Systems: SpringBoot, MyBatis, RocksDB, PostgreSQL, MySQL, Redis, Docker, Linux, Git

Languages Spoken: Mandarin Chinese (Native), English (Fluent)